FARI

Example Investigation Report

USB-Simulated Telecommand Interface

Release source: 02-usb-interface-report.md

FARI v1.2.0 | 2026-06-26

INCONCLUSIVE

Qualitative executive conclusion from the example report

This is the versioned release edition of the FARI documentation bundle. Canonical working sources for this release are stored under the same version folder.

Document Control

- **Assessment ID:** `ASM-FARI-2026-002`
- **Report version:** `0.2`
- **Report maturity:** Provisional
- **Report date:** `2026-06-07`
- **Source material:** `cases/02_USB.md` (../cases/02_USB.md)
- **Evidence producer:** Auditor; identity not supplied
- **Report author:** FARI Report Author
- **Mission owner / risk authority:** Not supplied
- **Framework profile:** SPARTA live content accessed `2026-06-07`

1. FARI Result Card

Field	Result
Conclusion	**Inconclusive**
Confidence	Low
Required Action	Extend Investigation
Priority	Planned before treating the USB interface as representative of the command path
Scope Boundary	Supplied USB test concept, script, assertions, and expected observations; no demonstrated command execution, measured effect, RF equivalence, or on-orbit behavior

Rationale: Relevant analysis and a credible hypothesis exist, but the supplied material contains no observed execution evidence and includes contradictory APID and payload descriptions. The claim cannot be resolved as Meets or Does Not Meet.

2. Executive Decision Summary

The supplied case describes a credible configuration-logic weakness: a `SET_BEACON_RATE` command may accept a zero-second interval, causing beacon generation to execute nearly continuously and potentially degrading telemetry visibility or other mission-loop functions.

The supplied source material does not include execution output, command acknowledgements, traffic captures, measured rate changes, source references, or recovery observations. It also contains an unresolved APID contradiction. Therefore, the weakness and mission effect are reported as **candidate**, not confirmed.

Decision required: Run a controlled, instrumented reproduction after resolving the APID and payload-format contradictions. Independently of the result, reject zero and unsafe beacon intervals in firmware before using the interface as a representative RF-path simulation.

3. Frame

Mission Context Added by the Report Author

The supplied source material does not identify a mission or architecture. For analysis, this report uses the following explicit assumptions:

- The USB-connected target is a FlatSat, engineering model, or representative spacecraft processor.
- Beacon telemetry shares processing, buffering, or link resources with normal mission telemetry.
- Maintaining readable, timely telemetry is a mission objective.

These assumptions require mission-owner confirmation.

Scope

Item	FARI Record
Primary asset	`AST-USB-TC-001` USB-exposed telecommand interface
Related asset	`AST-FSW-BEACON-001` Beacon-rate command handler and telemetry worker
Segment	Space segment representative environment and simulated link
Access basis	Private, as stated by evidence producer
Environment	USB serial at `921600` baud; dynamic lab execution claimed
Claim	`CLM-USB-001`: Configuration commands reject values that can cause uncontrolled telemetry generation or mission-loop degradation
Explicit exclusions	Actual RF transmission, on-orbit behavior, authentication path, full mission-loop resource analysis

Authorization

The case instructs use only during a controlled lab window, but no authorization record, stop condition, target revision, test date, or evidence-handling record was supplied.

4. Acquire: Source-Material Intake Review

Evidence-Backed Facts

- The supplied Python example constructs a CCSDS space-packet telecommand with `apid=6`, `seq_count=5`, and `data_len=0`.
- The example sends the packed header plus a newline to `/dev/cu.usbmodemfsat3` at `921600` baud.
- The execution narrative identifies target APID `0x05` and command `SET_BEACON_RATE`.

- The narrative defines a two-byte lab payload: interval plus padding.
- The narrative provides commands intended to send one-second, zero-second, and recovery intervals using `scripts/usb_tc_send.py`.
- The case contains expected observations but no recorded observations or command output.

Evidence-Producer Assertions

- USB is used to simulate the RF channel.
- The handler reads `data[0]`, rejects values above `10`, and accepts `0`.
- A zero interval makes the telemetry-worker condition true almost continuously.
- The result is a logic-level denial of service against the mission loop.

Report-Author Inferences

- If the handler and scheduling condition behave as described, accepting zero can create uncontrolled beacon generation.
- Uncontrolled beacon generation may consume processing or link capacity and reduce operator telemetry visibility.
- The provided case is a test plan and technical hypothesis rather than a completed execution record.

Contradictions and Quality Issues

ID	Issue	Reporting Decision
`GAP-USB-001`	Python example uses APID decimal `6`; execution narrative identifies APID `0x05` (decimal `5`)	Finding remains candidate until the correct APID is demonstrated
`GAP-USB-002`	Python example sends only a packed SPP header, while the reproduction requires interval and padding bytes	Require exact transmitted bytes and command acknowledgement
`GAP-USB-003`	Case declares static/offline and dynamic environments without identifying which evidence came from each	Report treats environment as an incompletely documented dynamic lab
`GAP-USB-004`	Final finding text refers to a library DoS in several packets, which does not match this USB beacon-rate scenario	Report excludes the copied finding statement
`GAP-USB-005`	Proposed `EX-0005.01` mapping is not the strongest match for the described software logic condition	Report uses candidate mappings to command packets, flight-software code flaws, and inhibited spacecraft functionality

Sufficiency Decision

Question	Decision	Rationale
Technical condition	Partial	Detailed producer assertion exists, but no source reference or observed execution output
External / operational reachability	Insufficient	USB delivery is described but not evidenced; RF equivalence and authentication are unknown
Mission consequence	Insufficient	Only expected effects are supplied; no measured degradation or recovery evidence

5. Relate: Candidate Finding and Mission Context

FND-USB-001: Zero Beacon Interval May Cause Uncontrolled Telemetry Generation

- **State:** Candidate
- **Condition:** The `SET_BEACON_RATE` handler is asserted to accept zero even though zero causes the scheduling condition to run nearly continuously.
- **Observed effect:** None supplied.
- **Expected effect:** Repeated beacon telemetry, increased USB/downlink traffic, and reduced readability of normal telemetry.
- **Preconditions:** Correct command APID and payload reach the handler; command is accepted; no upstream validation or authorization control rejects it.
- **Credible mission consequence:** Degraded telemetry visibility or mission-loop availability.
- **Technical confidence:** Low
- **Reachability confidence:** Low
- **Mission-consequence confidence:** Low

Framework Mappings

Mapping	State	Rationale
SPARTA `EX-0001.01 Command Packets` (https://sparta.aerospace.org/technique/EX-0001/01/)	Candidate	The scenario depends on delivery of a crafted but syntactically valid command packet
SPARTA `EX-0009.01 Flight Software` (https://sparta.aerospace.org/technique/EX-0009/01/)	Candidate	Accepting a resource-exhausting configuration value is a plausible flight-software logic flaw
SPARTA `DE-0002.03 Inhibit Spacecraft Functionality` (https://sparta.aerospace.org/technique/DE-0002/03/)	Candidate	Applicable if uncontrolled beacon activity is shown to inhibit normal telemetry generation or transmission
SPARTA `EX-0005.01 Design Flaws` (https://sparta.aerospace.org/technique/EX-0005/01/)	Not selected	No hardware, FPGA, ASIC, or firmware-corruption design flaw is demonstrated

Qualitative Mission Context

> Given command-path access, an actor or erroneous operator procedure could set the beacon interval to zero, potentially causing uncontrolled telemetry generation and degrading mission telemetry visibility or mission-loop availability.

The scenario could become operationally important if RF delivery is possible without strong command authentication, if zero persists across reset, or if beacon flooding interferes with command reception or safing. Those conditions were not demonstrated, so the appropriate executive result is to extend the investigation rather than assign a numeric risk value.

6. Inform

Required Controlled Reproduction

1. Confirm the intended APID and exact command identifier from the command dictionary.
2. Record the exact transmitted bytes for baseline, one-second, zero-second, and recovery commands.
3. Capture command acknowledgement or handler instrumentation proving command acceptance.
4. Measure baseline and test beacon rates, USB/link utilization, CPU usage, queue depth, dropped telemetry, and command responsiveness.
5. Record behavior during zero interval and recovery using predefined stop conditions.
6. Repeat after implementing validation that rejects zero and values outside an approved operational range.

Required Remediation

- Reject zero and unsafe beacon intervals before updating scheduler state.

- Define a mission-approved minimum and maximum interval.
- Use safe defaults when values are invalid or missing.
- Add command-level authorization and mode restrictions where appropriate.
- Rate-limit telemetry generation independently of configuration input.
- Alert operators when beacon rate changes or exceeds expected bounds.

Acceptance Criteria

- Zero and all out-of-range values are rejected with an explicit error.
- Rejected commands do not change the active interval.
- Beacon generation remains within the mission-approved rate under malformed, repeated, or unauthorized inputs.
- Normal telemetry and command responsiveness remain within defined thresholds.
- Recovery behavior is measured and documented.

Evidence Gaps and Follow-Up Requests

Gap ID	Requested Evidence	Why It Matters
`GAP-USB-006`	Correct APID, command dictionary entry, and exact wire bytes	Resolve contradictory targeting information
`GAP-USB-007`	Source reference for handler validation and scheduling condition	Confirm technical condition
`GAP-USB-008`	Execution logs, command acknowledgement, and traffic capture	Confirm command delivery and observed effect
`GAP-USB-009`	Baseline/test measurements and recovery observations	Quantify mission consequence
`GAP-USB-010`	Firmware revision, hardware revision, tool versions, and evidence hashes	Reproducibility
`GAP-USB-011`	RF-equivalence rationale and command-authentication controls	Evaluate operational reachability
`GAP-USB-012`	Mission telemetry thresholds and unacceptable outcomes	Final risk decision

Assurance Statement

`CLM-USB-001` is **Inconclusive** with low confidence. The supplied case describes a credible unsafe configuration condition, but it does not contain observed execution evidence. No conclusion is made about RF reachability, authentication bypass, measured mission degradation, persistence, or on-orbit behavior.

7. Coverage

Area	Coverage	Result
USB delivery concept	Reviewed	Script and procedure supplied; execution not evidenced
Beacon-rate handler logic	Inferred	Producer assertions supplied; source reference absent
Zero-interval behavior	Not assessed by supplied evidence	Expected observations only
Recovery	Not assessed by supplied evidence	Proposed procedure only
RF equivalence	Not assessed	No evidence supplied
On-orbit behavior	Out of scope	No evidence supplied

8. Closing FARI Result Card

Field	Result
Conclusion	**Inconclusive**
Confidence	Low
Required Action	Extend Investigation
Priority	Planned before representative-command-path acceptance
Next Decision	Resolve the claim after controlled execution captures the exact command, observed behavior, and recovery